Rajalakshmi Engineering College

FUNDAMENTALS OF DATA SCIENCE LAB

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EXPERIMENT 7

Random sampling and sampling distribution

Experiment 7: Random sampling and sampling distribution

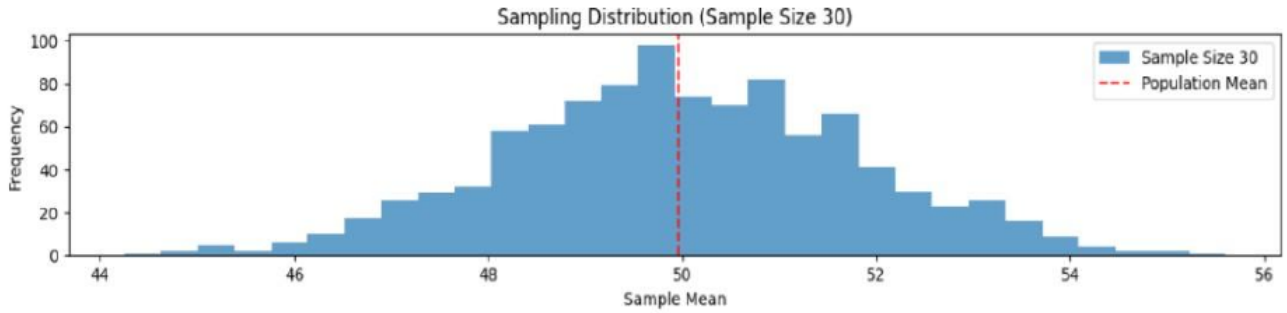
Aim

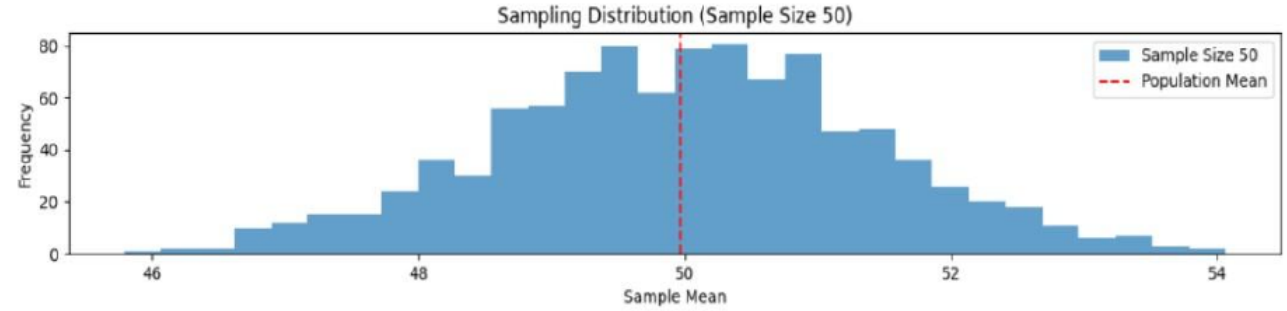
To demonstrate random sampling and the sampling distribution of the sample mean.

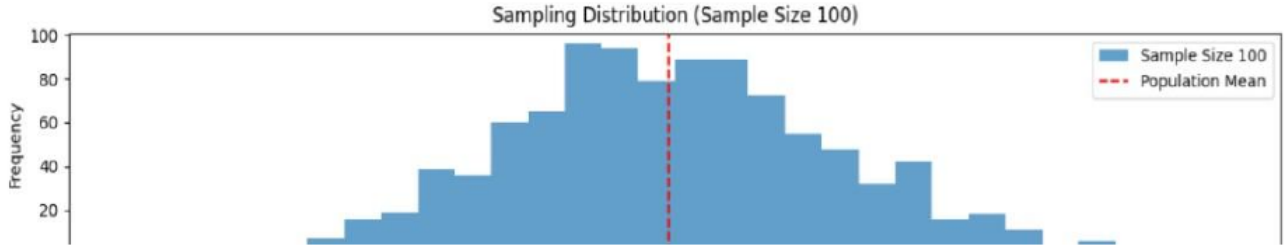
Source Code

```
import numpy as np
import matplotlib.pyplot as plt
population_mean = 50
population_std = 10
population_size = 100000
population = np.random.normal(population_mean, population_std, population_size)
sample_sizes = [30, 50, 100]
num_samples = 1000
sample_means = {}
for size in sample_sizes:
sample_means[size] = []
for _ in range(num_samples):
sample = np.random.choice(population, size=size, replace=False)
sample_means[size].append(np.mean(sample))
plt.figure(figsize=(12, 8))
for i, size in enumerate(sample_sizes):
plt.subplot(len(sample_sizes), 1, i+1)
plt.hist(sample_means[size], bins=30, alpha=0.7, label=f'Sample Size {size}')
plt.axvline(np.mean(population), color='red', linestyle='dashed', linewidth=1.5, label='Population
Mean')
plt.title(f'Sampling Distribution (Sample Size {size})')
plt.xlabel('Sample Mean')
plt.ylabel('Frequency')
plt.legend()
plt.tight_layout()
plt.show()
```

Output - Experiment 7







Output - Experiment 7

Experiment 7:. Random Sampling and Sampling Distribution
Objective:
To explore random sampling from a population and understand the concept of sampling distribution
Steps:
1. Generate a Population:
○ Create a population of data with a specified distribution (e.g., normal distribution).
2. Random Sampling:
○ Perform random sampling from the population to create multiple samples of
different sizes.
○ Compute sample statistics (mean, standard deviation, etc.) for each sample.
3. Sampling Distribution:
○ Plot histograms or density plots of sample statistics (e.g., sample means).
○ Compare the sampling distribution of the sample statistic (mean) with the
population distribution.
4. Central Limit Theorem (Optional):
○ Demonstrate the Central Limit Theorem by showing that as sample size increases,
the sampling distribution of the sample mean approaches a normal distribution
regardless of the population distribution.
Example Code:
python

Result - Experiment 7

Random sampling and sampling distribution were demonstrated.